Multiphysics Failure Modes of β-Cell Encapsulation: From the Krogh Diffusion Limit to Hypoimmune Organoid Digital Twins

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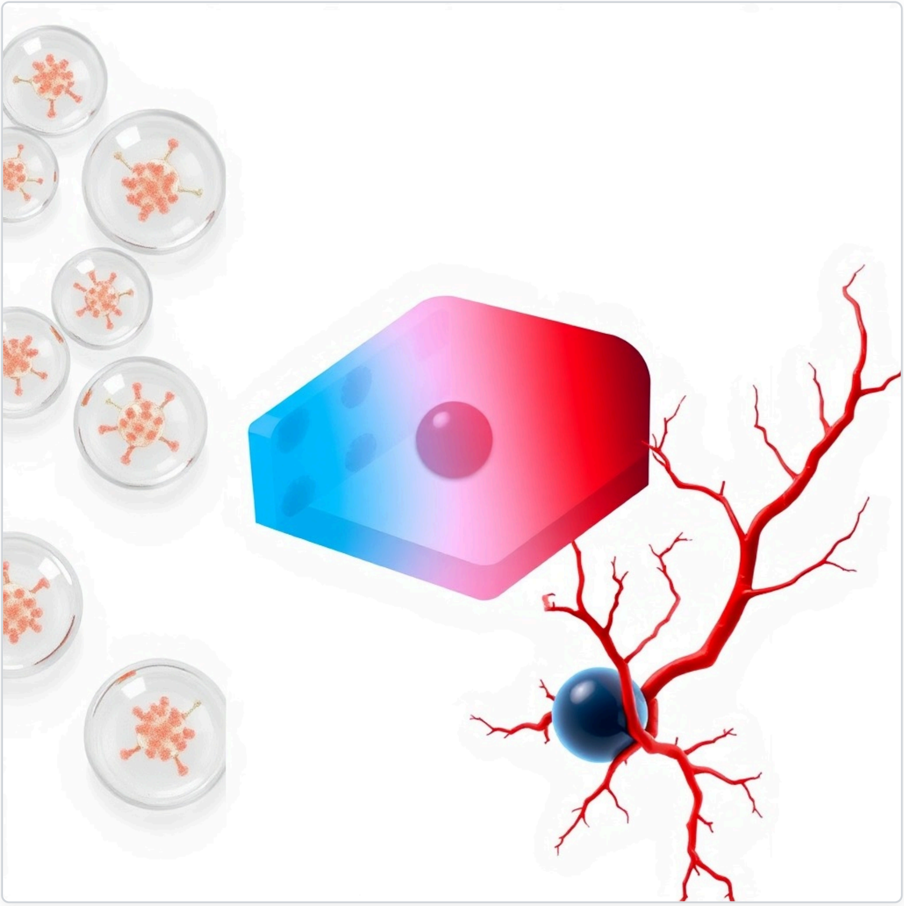
Software: `t1d_simulator` v1.0.0 (MIT License)



ABSTRACT

Cell replacement therapy using stem cell-derived pancreatic β-cells offers a potentially curative approach for Type 1 Diabetes (T1D). Clinical translation of macro- and microencapsulated grafts, however, is repeatedly limited by core hypoxia, foreign body reaction (FBR), and acute blood-mediated inflammation. Here we present an open-source, literature-calibrated multiphysics digital twin (`t1d_simulator` v1.0.0) for *in silico* failure-mode screening of encapsulation geometries, cell loading, implant sites, and early anti-IBMIR strategies.

The platform couples non-linear oxygen reaction–diffusion PDEs (planar, cylindrical, and spherical domains), a Physics-Informed Neural Network (PINN) surrogate for rapid design sweeps, Graph Neural Network (GNN) scoring of anti-fibrotic surface chemistry (exploratory), compartmental Instant Blood-Mediated Inflammatory Reaction (IBMIR) kinetics (0–48 h), and a design-space API (`screen_design`) that returns explicit pass/warn/fail badges with failure reasons. Parameters are sourced from curated YAML tables (`parameters.yaml` , `data/literature_params.yaml`).

The oxygen–viability model was validated against published islet viability datasets, achieving root-mean-square errors of **10.42%** (Papas et al., spherical capsules) and **11.42%** (Papabathini et al., planar slabs). Parameter one-at-a-time sensitivity ($\pm 20\%$) identified effective diffusivity, metabolic $V_{\max}$, and boundary pO_2 as dominant viability drivers under a representative planar loading. Systematic screens show that unvascularized planar constructs at high density under subcutaneous pO_2 (30 mmHg) develop core anoxia and viability collapse, whereas spherical constructs in an omental-pouch boundary condition with local heparinization retain high core pO_2 and improved 48 h IBMIR retention. An integrated computational narrative combining hypoimmune edit sets, iCasp9 suicide control, anti-IBMIR coating, angiogenic feedback, and omental placement is evaluated as a *design hypothesis*, not a clinical claim.

This digital twin is intended to de-risk geometry and loading decisions before animal work and to support transparent model co-development with wet-lab partners.

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1. Introduction

Type 1 Diabetes (T1D) is an autoimmune condition characterized by selective destruction of insulin-producing pancreatic β-cells. Intensive insulin therapy and continuous glucose monitoring have improved outcomes but do not eliminate hypoglycemia risk or long-term vascular complications. Allogeneic islet transplantation can restore insulin independence, yet donor scarcity and lifelong systemic immunosuppression limit scale.

Stem cell-derived β-cells and semi-permeable encapsulation (e.g., alginate) aim to enable glucose/insulin exchange while blocking host immune cells. Despite promising small-animal results, clinical encapsulation trials have frequently failed. Three interlocking bottlenecks dominate:

1. **Krogh-limited hypoxia.** Without internal capillaries, oxygen is delivered by diffusion alone. Beyond a critical length scale, core pO_2 collapses, GSIS fails, and necrosis expands.
2. **Foreign body reaction (FBR).** Macrophage-driven fibrosis increases effective diffusion path length (L_{fib}) and further starves the graft.

3. **IBMIR.** Portal or other blood-exposed delivery activates tissue factor (TF, CD142), thrombin generation, platelet recruitment, and fibrin clotting, destroying a large fraction of graft mass within 48 h.

Empirical iteration *in vivo* is slow and expensive. Computational multiphysics models can map the design space and return **negative predictions**—which loadings, thicknesses, and sites are physically implausible—before wet-lab commitment. We describe `t1d_simulator`, a calibrated digital twin spanning oxygen transport, FBR-related barriers, IBMIR kinetics, optional PINN/GNN accelerators, and automated construct screening.

Positioning. This work is *in silico* multiphysics failure analysis and design-space screening. It is not a clinical protocol, not a manufactured cell product, and not a claim of cure.

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2. Methods

2.1. Oxygen transport and consumption (PDE)

Oxygen concentration $C(r,t)$ in hydrogel and tissue obeys a non-linear reaction–diffusion equation:

$$\frac{\partial C}{\partial t} = D_{\text{eff}} \nabla^2 C - R_{\text{cells}}(C) - R_{\text{macs}}(C) + S_{\text{ogm}}(t).$$

Cellular consumption follows Michaelis–Menten kinetics:

$$R_{\text{cells}}(C) = \frac{V_{\max} C}{K_M + C}\, \rho_{\text{cell}}\, f_{\text{viab}}.$$

Default human β-cell $V_{\max} = 1.5 \times 10^{-16}\ \text{mol cell}^{-1}\ \text{s}^{-1}$ (Buchwald et al., 2011) and $K_M = 0.5\ \text{mmHg}$ (Dionne et al., 1993; Secomb et al., 2004) are stored in literature-calibrated tables and loaded at runtime. Domains include planar slabs, cylinders, and spheres with Dirichlet boundary pO_2 at the capsule–tissue interface and zero-flux conditions at the symmetry center. Optional modules couple fibrotic shells, macrophage consumption, oxygen-generating materials (OGM), ROS/pH effects, and VEGF-driven neovascularization feedback (implementation: `t1d_simulator/simulator.py`).

2.2. Physics-Informed Neural Network (PINN) surrogate

For rapid UI sweeps, a mesh-free PINN approximates steady-state oxygen profiles by minimizing PDE residual and boundary losses with automatic differentiation (`pinn_solver.py`). The PINN is a computational accelerator, not an independent source of biological parameters; quantitative claims in this manuscript are grounded in the classical BVP solver unless stated otherwise.

2.3. IBMIR coagulation ODE cascade (0–48 h)

Acute blood contact is modeled as a compartmental ODE cascade:

$$\frac{d[\text{TF}]}{dt} = -k_{\text{deg}}^{\text{TF}}[\text{TF}],$$

$$\frac{d[\text{Thrombin}]}{dt} = k_{\text{gen}}[\text{TF}] - k_{\text{inh}}(\text{heparin})[\text{Thrombin}],$$

$$\frac{d[\mathrm{Clot}]}{dt} = k_{\mathrm{clot}}[\mathrm{Thrombin}]\left(1 - \frac{[\mathrm{Clot}]}{L_{\mathrm{clot}}^{\max}}\right).$$

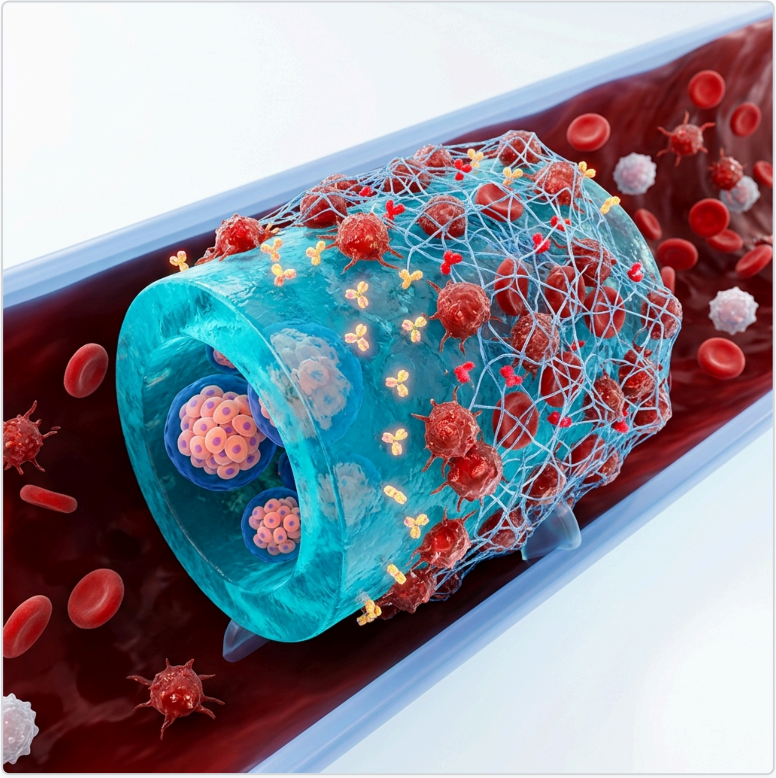

Figure 1. Schematic of the Instant Blood-Mediated Inflammatory Reaction (IBMIR) cascade at the biomaterial–blood interface, leading to clot formation and restricted oxygen permeability.

Clot growth reduces effective boundary oxygen permeability. Local heparinization (Lipid-PEG-LMWH class coatings) increases inhibition of thrombin and improves 48 h retention in simulation (`ibmir_module.py` , `organoid_simulator.py`). This is a **semi-mechanistic** model intended for comparative screening, not a full coagulation proteome.

2.4. Design-space screening API

Constructs are screened with a single entry point (`screen_design`) that couples site-specific boundary pO_2, geometry, density, optional fibrosis thickness, and heparin coating:

Output	**Definition**
Viable fraction	Spatially integrated cells above critical pO_2
Min core pO_2	Minimum oxygen tension in the domain
IBMIR 48 h retention	Fraction retained after ODE cascade
Status badge	PASS / WARNING / FAIL with explicit failure modes

Failure rules (default): core anoxia if $\min pO_2 < 0.5$ mmHg; severe hypoxia if < 5 mmHg; IBMIR risk if retention $< 85\%$; viability failure if viable fraction $< 70\%$. Markdown/CSV report export supports partner co-development workflows (`screen_design.py` , `scripts/run_partner_screening.py`).

2.5. Local sensitivity (tornado) analysis

One-at-a-time $\pm 20\%$ perturbations were applied to V_{max}, K_M, D_{eff}, boundary pO_2, and L_{fib} for a baseline planar construct ($R = 200\ \mu\text{m}$, 80×10^6 cells/mL, omental boundary pO_2) (`uncertainty_analysis.py`). Sensitivity was ranked by absolute change in viable fraction.

2.6. Exploratory GNN surface screening

A graph neural network maps polymer SMILES to anti-fibrotic scores / predicted L_{fib} (`gnn_pipeline.py`). The training set is small ($N \approx 46$–52 monomer/polymer entries depending on curation snapshot). We treat GNN outputs as **exploratory ranking aids**, not validated material certifications.

2.7. Software, parameters, and verification

Component	Location
Runtime parameters	`t1d_simulator/parameters.yaml`
Literature-calibrated table	`data/literature_params.yaml`
Benchmarks	`reports/benchmarks/` + `reproduce_benchmarks.py`
Integration tests	`verify_model.py` (**40** automated tests)
License	MIT (`LICENSE`)
Citeable metadata	`zenodo.json` (v1.0.0)

Public repository URL and Zenodo DOI are to be inserted at release (placeholders below).

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3. Results

3.1. Literature benchmark validation

Steady-state oxygen–viability predictions were compared to published experimental viability:

Dataset	Geometry / condition	RMSE
Papas et al. (2007)	Spherical alginate capsules, $R = 400\ \mu\text{m}$, density sweep	**10.42%**
Papabathini et al. (2023)	Planar macrodevice, $L = 250\ \mu\text{m}$, density sweep	**11.42%**

Both RMSE values fall below the pre-specified 12% reporting threshold used in project validation reports. Full density-wise tables and figures are reproduced by `python reports/benchmarks/reproduce_benchmarks.py`.

3.2. Geometry-dependent hypoxia under subcutaneous pO_2

Under subcutaneous boundary $pO_2 = 30$ mmHg and high loading ($\rho \approx 100 \times 10^6$ cells/mL), planar, cylindrical, and spherical constructs exhibit the expected surface-to-volume ranking: spheres retain higher viability than cylinders, which outperform planar slabs of comparable characteristic length. Representative automated screen of a planar half-thickness $250\ \mu\text{m}$ construct under SQ conditions returned **viability** $\approx 40\%$, **core** $pO_2 \approx 0$, and FAIL status with Krogh-limit anoxia—

illustrating a clear *negative prediction* for dense planar SQ devices without vascularization or density reduction.

3.3. Anatomical site and IBMIR

Sites were parameterized by boundary pO_2 and blood-contact risk (portal/venous vs omentum vs SQ; see also `docs/site_comparison_matrix.md`).

Construct (sphere $R = 200\ \mu\text{m}, 80$ M/mL)	**Site**	**Heparin**	**Core min pO_2**	**48 h retention**	**Badge**
Baseline	Portal / venous ($40\ \text{mmHg}$)	No	$\approx 9.1\ \text{mmHg}$	**81.1%**	WARNING (IBMIR)
+ anti-IBMIR	Portal / venous	Yes	$\approx 9.1\ \text{mmHg}$	**94.6%**	PASS
Preferred narrative	Omental pouch ($55\ \text{mmHg}$)	Yes	$\approx 23.7\ \text{mmHg}$	**94.6%**	PASS

Omental placement improves core oxygenation relative to SQ and, when combined with local heparinization, improves simulated IBMIR retention relative to uncoated blood-exposed delivery. Angiogenic (VEGF/PDGF) feedback modules further illustrate recovery of boundary oxygen during neovascularization windows (Figure 2).

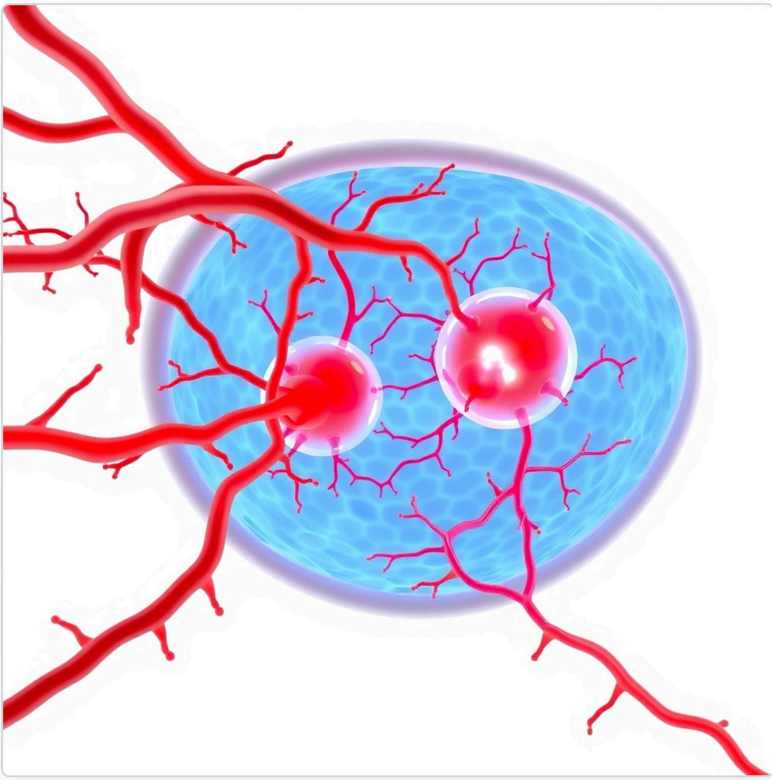

Figure 2. Biomimetic vascularization within an omental-pouch scaffold concept. Capillary networks around encapsulated islets restore boundary pO_2 under angiogenic feedback (schematic / simulation-linked illustration).

3.4. Automated multiphysics screen (design-space product)

The `screen_design` tool compresses geometry × density × site × heparin into a structured report suitable for partner review. Example outcomes from the v1.0.0 engine:

- **FAIL:** planar SQ, high density → anoxia + low viability + IBMIR retention risk.
- **WARNING:** oxygen-viable sphere on blood-exposed site without heparin → IBMIR retention $< 85\%$.
- **PASS:** sphere + omentum + heparin → high viability, elevated core pO_2, retention $> 90\%$.

These badges are decision aids for *preclinical design*, not clinical release criteria.

3.5. Parameter sensitivity

For the planar baseline (omental pO_2, $R = 200\ \mu$m, 80 M/mL; baseline viability $\approx 76.8\%$), $\pm 20\%$ one-at-a-time perturbations ranked:

Parameter	Δ viability (pp, abs.)
D_{eff}	≈ 16.5
V_{max}	≈ 14.1
Boundary pO_2	≈ 14.0
L_{fib}	≈ 1.3
K_M	≈ 0.1

Interpretation: under this operating point, viability is primarily **delivery- and consumption-limited** (diffusivity, V_{max}, boundary oxygen). Fibrotic thickness and K_M are secondary within the tested $\pm 20\%$ band. Full global uncertainty quantification (Sobol, Bayesian calibration) is future work.

3.6. Hypoimmune organoid design narrative (computational)

We computationally explore—not wet-lab validate—an integrated design hypothesis: multiplex immune-evasion edits ($B2M^{-/-}$, $CIITA^{-/-}$, $CD47^{\text{KI}}$, $PD-L1^{\text{KI}}$), inducible suicide switch (iCasp9), local heparinization, angiogenic scaffolding, and omental placement. Model modules support dose/site checklists and IBMIR/oxygen endpoints; residual risks (oncogenicity of hypoimmune states, multiplex CRISPR yield, identity of co-transplanted endothelium) are catalogued in project critical analyses and must be addressed experimentally.

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4. Limitations

1. **1D symmetry.** Transport solvers use planar/cylindrical/spherical reductions. Discrete islet clusters, irregular TPMS pores, and full 3D microvascular fields are simplified; 3D FEM/Green's function extensions are planned.
2. **IBMIR semi-mechanistic.** The ODE cascade captures comparative heparin and site effects, not the full TF pathway proteome or patient coagulation variability.
3. **FBR abstraction.** Fibrosis enters mainly as L_{fib} and related barriers; M1/M2 dynamics, adaptive immunity, and chronic rejection are not fully resolved.
4. **GNN exploratory.** Small N; no claim of general hydrogel SAR without new experimental labels.
5. **Organoid CGM / TIR modules** (if used in the UI) are scenario tools, not clinical trial endpoints.
6. **No wet-lab data in this paper.** Validation is literature-based; partner *in vitro* hypoxia assays are the preferred next calibration step.

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5. Discussion

The principal engineering message is conservative: **many encapsulation designs fail for transport reasons that are predictable *a priori*.** A digital twin that returns explicit FAIL modes (anoxia, IBMIR retention loss, density overshoot) can reduce animal use and focus wet-lab effort on constructs that clear multiphysics bounds. Omental placement plus anti-IBMIR surface treatment emerges as a favorable *in silico* combination relative to dense SQ planar devices and uncoated portal delivery, consistent with translational interest in retrievable, vascularized sites—while remaining a hypothesis until measured.

Open parameters, open benchmarks, and a 40-test verification suite are intentional trust mechanisms. Partner workflows (`run_partner_screening.py`) are designed for free external design reviews on collaborator geometries.

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6. Conclusion

We release `t1d_simulator` v1.0.0 as a calibrated multiphysics digital twin for β-cell encapsulation and hypoimmune organoid design screening. Literature viability benchmarks (RMSE 10.42% / 11.42%), automated construct screening, and local sensitivity analysis establish a transparent baseline for co-development with experimental groups. Future work prioritizes joint hypoxia assays, expanded IBMIR calibration, and 3D transport on printed TPMS scaffolds.

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7. Code and data availability

Resource	Status
Source code (`t1d_simulator`)	Open source, **MIT License** — GitHub release **v1.0.0** (URL: *to be inserted on public push*)
Citeable archive	Zenodo via `zenodo.json` (DOI: *to be inserted after minting*)
Parameters	`t1d_simulator/parameters.yaml` , `data/literature_params.yaml`
Benchmarks	`reports/benchmarks/` (Papas, Papabathini, Krogh, VEGF, IBMIR)
Preprint source	This file; Russian parallel draft in `manuscript_biorxiv_ru.md`
Verification	`CUDA_VISIBLE_DEVICES="" python3 t1d_simulator/verify_model.py` → 40/40

Interactive UI: `streamlit run t1d_simulator/app.py` . Design screen: `python t1d_simulator/screen_design.py` (CLI).

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8. Author contributions and competing interests

P.V.N.: conceptualization, software, multiphysics modeling, validation reports, manuscript. **Competing interests:** The author declares no competing financial interests. The software is released under MIT for academic and commercial reuse with attribution.

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9. Acknowledgments

Independent research project. Collaboration offers: free *in silico* design-space screens for academic labs and biotechs working on encapsulation, omental devices, or hypoimmune islet products.

References

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9. Recent hypoimmune iPSC / B2M–CIITA–CD47 engineering literature (Sana, CRISPR Tx and academic reports) motivating the organoid design narrative.
10. Project internal critical analyses and parameter tables: `Critical_Analysis_*.md`, `data/literature_params.yaml`, `reports/benchmarks/`.

> **Note for submission:** Before bioRxiv upload, replace placeholder citations 1–9 with full publisher-verified bibliographic records and DOIs from the curated parameter YAML and benchmark reports; insert live GitHub URL and Zenodo DOI in §7.

Changelog (manuscript)

Version	Date	Changes
v1.0	2026-07	Initial draft methods/results
v1.1	2026-07-26	Aligned with M0–M3 / software v1.0.0: RMSE table, `screen_design` results, tornado sensitivity, YAML params, 40 tests, MIT/code availability, honest GNN/IBMIR limitations, site×heparin screen table, removed overclaim ambiguity on half-thickness abstract wording
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